

High dietary intake of saturated fat is associated with reduced semen quality among 701 young Danish men from the general population.

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BACKGROUND: Saturated fat intake has been associated with both cardiovascular disease and cancer risk, and a newly published study found an association between saturated fat intake and a lower sperm concentration in infertile men. OBJECTIVE: The objective was to examine the association between dietary fat intake and semen quality among 701 young Danish men from the general population. DESIGN: In this cross-sectional study, men were recruited when they were examined to determine their fitness for military service from 2008 to 2010. They delivered a semen sample, underwent a physical examination, and answered a questionnaire comprising a quantitative food-frequency questionnaire to assess food and nutrient intakes. Multiple linear regression analyses were performed with semen variables as outcomes and dietary fat intakes as exposure variables, adjusted for confounders. RESULTS: A lower sperm concentration and total sperm count in men with a high intake of saturated fat was found. A significant dose-response association was found, and men in the highest quartile of saturated fat intake had a 38% (95% CI: 0.1%, 61%) lower sperm concentration and a 41% (95% CI: 4%, 64%) lower total sperm count than did men in the lowest quartile. No association between semen quality and intake of other types of fat was found. CONCLUSIONS: Our findings are of potentially great public interest, because changes in diet over the past decades may be part of the explanation for the recently reported high frequency of subnormal human sperm counts. A reduction in saturated fat intake may be beneficial for both general and reproductive health.